Gradient boosting for classification

	Chest pain	Good Blood Circulation	Blocked arteries	Heart disease		Residual
				prob of y=0		
S_1	NO	NO	NO	NO	0	0-0.66
S_2	Yes	Yes	Yes	Yes	1	1-0.66
S_3	Yes	Yes	NO	Yes	1	"
S_4	Yes	NO	NO	Yes	1	"
S_5	Yes	NO	Yes	Yes	1	"
S_6	NO	Yes	NO	NO	0	0-0.66

→ instead of $f(x;\theta)=0$ → log odds
→ convert to prob

Step 1: Initial prediction

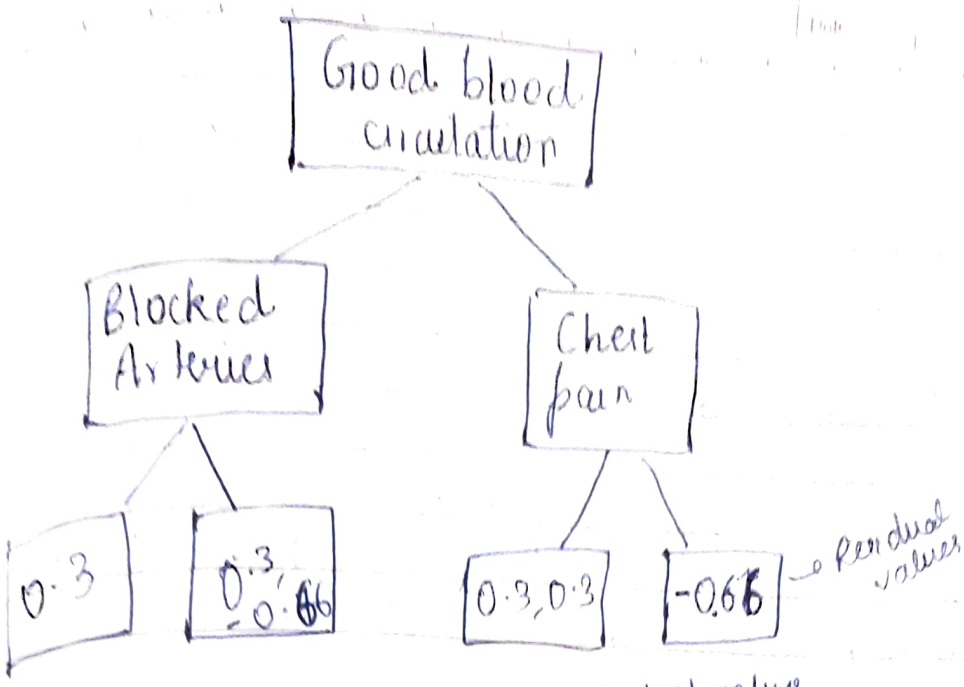
i) $\log(\text{odds}) = \ln\left(\frac{4}{2}\right) \approx 0.7$
↳ for Yes

ii) Convert this to prob.

$$p = \frac{e^{\log(\text{odds})}}{1+e^{\log(\text{odds})}}$$ → to convert betn 1 & 0

$$= \frac{e^{\log\left(\frac{4}{2}\right)}}{1+e^{\log\left(\frac{4}{2}\right)}} = 0.66$$

ln?



Predict residuals & convert the leaf values to log (odds)

$$\frac{\Sigma \text{Residuals}_i}{\Sigma\left[\text{Previous Prob}_i \times (1 - \text{Prev Prob}_i)\right]} \Rightarrow \log(\text{odds})$$

$$1^{st} \text{ leaf} = \frac{0.3}{(0.66 \times (1 - 0.66))} \approx 1.43$$

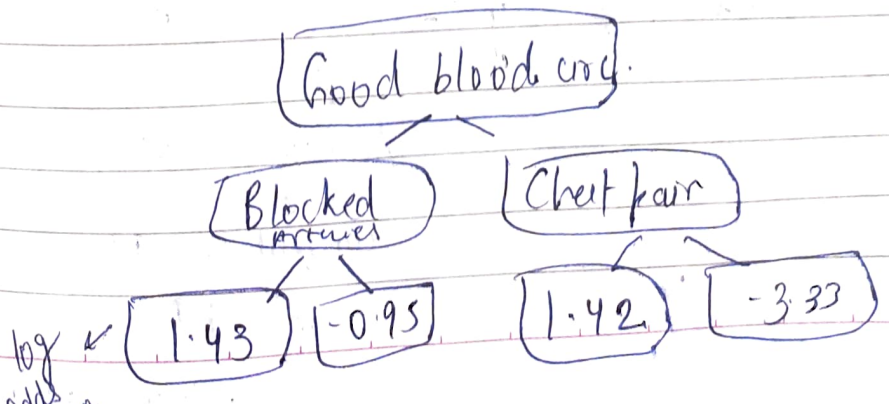
$$\text{for } 2^{nd} \text{ leaf} \Rightarrow \frac{0.3 + (-0.7)}{(0.76 \times (1 - 0.76)) + (0.76 \times (1 - 0.76))}$$

$0.66 \approx 0.7$

$$= \frac{-0.4}{0.4488}$$

≈ 0.42

$$= -0.95$$



convert to prob

$$\Rightarrow \frac{e^{\ln(1.43)}}{1+e^{\ln(1.43)}} = 0.588 \approx 0.6$$